EXPERIMENT - 2

Stages of C Compilation and Execution

1.Objective

To demonstrate the stages of C program compilation using the -save-temps flag and to write, compile, and execute a C program that calculates the area of a circle.

2. Theory: Stages of C Compilation

The process of converting a C source code file into an executable binary involves four primary stages:

2.1 Preprocessing

The preprocessor removes comments, expands macros (like #define), and includes the contents of header files (like stdio.h). It generates an intermediate .i file.

2.2 Compilation

The compiler translates the preprocessed C code into assembly language instructions specific to the target architecture. It generates a .s file.

2.3 Assembly

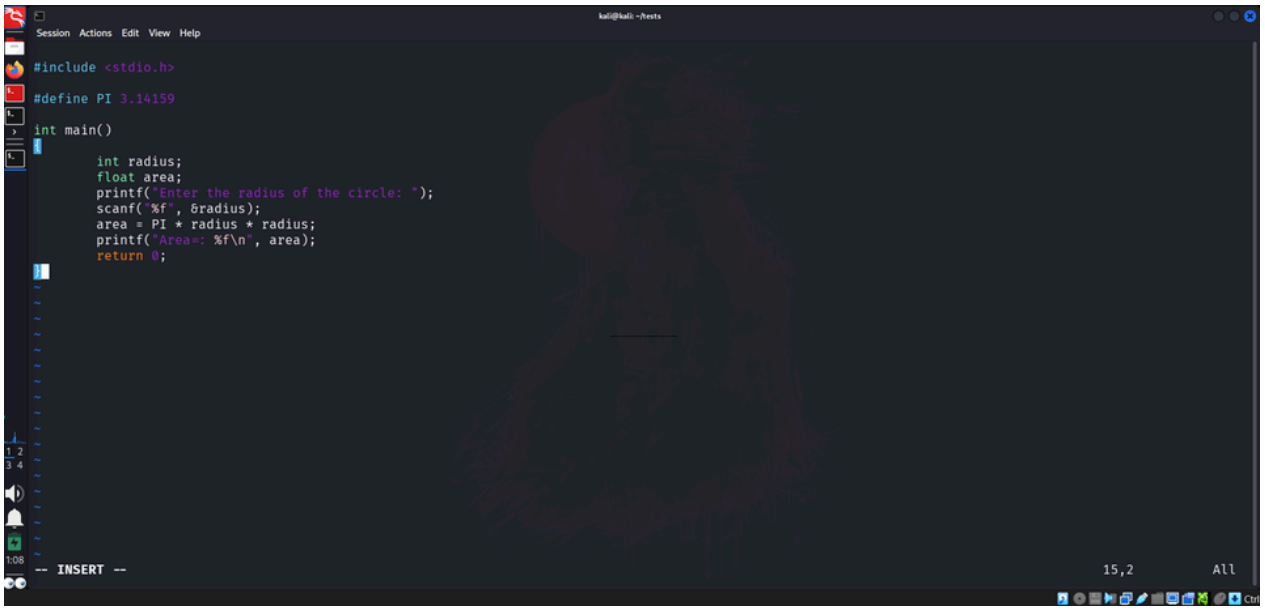
The assembler converts the assembly language code into machine-level object code (binary). It generates a .o file.

2.4 Linking

The linker combines the generated object code with standard library files to resolve function calls (e.g., printf, scanf) and creates the final executable file

3. The C Program: Area of a Circle

The following C program calculates the area of a circle using the standard formula Area = πr2



4. Demonstrating Compilation Stages

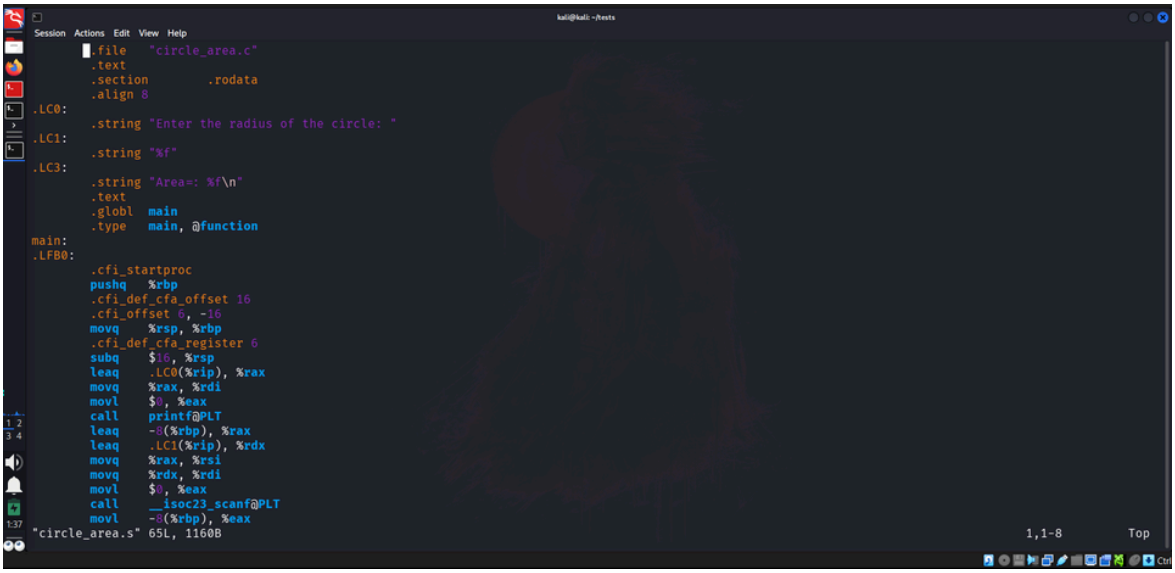
To observe all the intermediate files generated during the compilation process, we use the -save-temps flag with cc compiler.

4.1 Compiling with -save-temps

4.2 Analysis of Intermediate Files

- circle_area.i (Preprocessed Code): Opening this file reveals that the code has expanded significantly due to the inclusion of stdio.h. Furthermore, the comment has been removed, and the PI macro inside the calculation has been explicitly replaced with 3.14159.

- circle_area.s (Assembly Code): This file contains the assembly-level instructions generated by the compiler. Below is a captured view of the assembly file content:

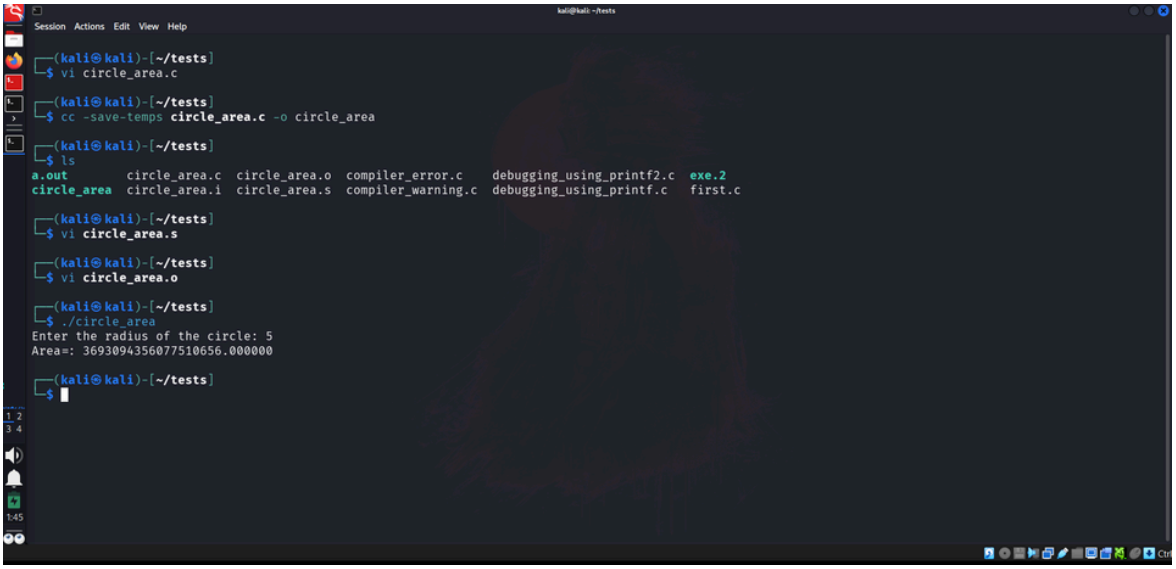


- circle_area.o (Object Code / Machine Code): This contains raw binary machine code. Attempting to view it with a standard text editor yields unreadable characters. To inspect it, a hexadecimal dump utility like hexdump is used. Below is a captured hex dump of the machine code:

- circle_area (Executable): The final linked binary file ready for execution by the operating system.

5. Program Execution

Executing the final compiled binary to calculate the area.



6. Conclusion:

Conclusion: The stages of compilation were successfully analyzed using intermediate files, and the logic to calculate the area of a circle was executed properly.